A system for named entity recognition based on local grammars

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Abstract

The existence of large-scale lexical resources for Serbian, e-dictionaries in particular, coupled with local grammars in the form of finite-state transducers, enabled the development of a complex system for named entity recognition and tagging. The system is not general in nature, but targets some specific types of name, temporal and numerical expressions. In order to improve the precision of recognition we used local grammars to describe the context of named entities. In the case of personal names the widest context was used to include the recognition of nominal phrases describing a person's position. The evaluation of our system was performed twice on a corpus of 3,000 short agency news. Results obtained by the system were manually evaluated, all omissions and incorrect recognitions precisely identified, and most of them corrected before the second evaluation. The overall recall $R=0.88$ for types and $R=0.94$ for tokens, and overall precision $P=0.96$ for types and $P=0.98$ for tokens indicated that our system gives priority to precision. The evaluation of recognition of surnames only, with and without positions, and also names of distinguished persons such as royalty and church dignitaries confirmed this fact, albeit with less satisfactory results for both precision and recall.

Keywords: Lexical resources, finite-state transducers, local grammars, named entity recognition, Serbian language, system evaluation.

1 Introduction

Recognition of named entities (NER) has been a hot topic in Natural Language Processing community for more than 15 years. Ever since their introduction in the scope of the Sixth Message Understanding Conference [9] they have not ceased to arouse interest of developers of various NLP applications. In [20] Przepiórkowski gives a good state-of-the-art review of Information Extraction (IE) for Slavic Languages, and analyses why Slavic languages are difficult to process, but also gives some arguments in their favour. The author explains how Slavic rich morphology can help at higher levels of processing, such as parsing. Hence, as shallow parsing is in general often successfully used for named entity recognition (as will be demonstrated for Serbian in the following sections), it is to be expected that it is even more often used for NER in Slavic languages.

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Published online February 19, 2013 doi:10.1093/logcom/exs079

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Evidence from recently produced Language White Papers in the scope of META-NET project supports such a conclusion [21]. Namely, in each of 32 languages for which such a white paper was produced, the state of language technology, divided in 13 groups, was measured on a scale from 0 to 6 for quantity, availability, quality, coverage, maturity, sustainability and adaptability. One of this groups covered IE technology including named entity recognition, event/relation extraction, opinion/sentiment recognition and text mining/analytics. It is interesting to note that for seven Slavic languages for which white papers were written (Bulgarian [1], Croatian [24], Czech [2], Polish [17], Serbian [26], Slovak [27] and Slovene [13])[1] average scores where reasonably high within the IE group. They range from 1.71 for Slovak to 3.29 for Czech (for Serbian 2.14) thus being at the similar level as scores for English, German (both having a score of 2.71) and French (3.29). Such good scores are most probably due to NER systems, and they are even more striking in view of the fact that for all of these languages language technology from some other major groups is non-existent. The number of these missing groups for Slavic languages ranges from two in the case of Bulgarian and Czech to as many as 6 for Slovak (5 for Serbian). Needless to say, for English, German and French technologies from all groups exist, at least to a certain degree. It can, therefore, be concluded that NER for Slavic languages is at a satisfactory level.

Gross shows in [10] that finite-state automata can be successfully used for lexical representation of various language phenomena, like variations of spelling, derivational morphology and syntactic structures, exemplified by French preverbal particles and frozen sentences. He defines in [12] local grammars as 'finite-states grammars or finite-state automata that represent sets of utterances of a natural language'. In [11] Gross advocates for the development of grammars of a local nature as opposed to those that 'are driven by a quest for complete generalizations'. In this approach the interaction of a multiplicity of local grammars or local finite-state schemes enables the description of the global nature of a language. The advantage of this approach is that finite constraints can be exhaustively described by one local grammar without interferences from the rest of the grammar (other local grammars).

An example given in [11] illustrates another characteristic of this approach: phrases such as *English speaking student* can be described by a local grammar *<N+Lang> speaking (<N+Hum>+<N+HumColl>)*[2], meaning that the position of *English* can be filled by the name of any language whereas the position of *student* can be filled by any human noun, representing either an individual or a group. This example shows how local grammars can be successfully coupled with lexicons in which entries have been marked with various features. In [12] a procedure is described for step-by-step construction of local grammars: existing lexical resources and grammars are used to build a new local grammar, upon which the new local grammar is tested on a corpus, and on the basis of obtained results both existing resources and the new local grammar are improved and then tested again.

The nature of proper names, as a sub-class of named entities, for Serbian was analysed in [25] especially in connection with its inflectional and derivational richness. However, to the best of our knowledge, beside some small-scale experiments, no effective NER system was yet produced for Serbian. In this article we present a working system for recognition of various named entities in Serbian newspaper texts, which is based on the concept of local grammars introduced by Maurice Gross and on his guidelines for building them. In building our local grammars we use the full strength of grammars implemented in the Unitex corpus processing system [19]. At the end we give results of the evaluation of our NER system on a corpus of short agency news.

[1]Language White Papers for Bulgarian, Croatian, Polish, Serbian and Slovak were produced in the scope of the ICT-PSP project CESAR (Central and South-East European Resources), contract no. 271022.

[2]This local grammar is written in Unitex formalism (http://www-igm.univ-mlv.fr/~unitex).

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TABLE 1. The system of Serbian e-dictionaries

Sub-dictionaries	Simple words		Compounds	
	Lemmas	Forms (K)	Lemmas	Forms (K)
General lexica	91,281	3,899	8,423	185
Geopolitical proper names	8,032	217	879	9
Serbian personal names	20,781	275		
Foreign proper names	6,688	47		
Encyclopedic knowledge	1,253	23	395	4
Total	128,035	4,461	9,697	198

2 General resources used for the NER task

The primary resources that we have used for the NER task are Serbian morphological e-dictionaries. The system of Serbian e-dictionaries covers both general lexica and proper names, as simple words and compounds, and their level of development is presented in Table 1. It is obvious from the given data that dictionaries of compounds are still of a modest size and need to be further developed. On the other hand, they comprise not only entries collected from traditional sources, like dictionaries, but also entries extracted from processed texts, which enhances their usability.

The development of our e-dictionaries follows the methodology and format known as DELA presented for French in [5]. Their content is based on the precise and thorough classification of the inflectional properties of words, being simple or multi-word units. To that end, every entry in a dictionary of lemmas is associated with a code describing its part-of-speech and possibly inflectional, syntactic and semantic properties. Inflectional properties are represented by an inflectional class code that describes inflectional endings and the grammatical meaning assigned to them. For example, the information associated with the lemma:

$$\textit{Ženevljanin}, N60+NProp+Hum+Inh+CH$$

states that *Ženevljanin* is a noun (N) 'a citizen of Geneva', which is a proper name (+NProp), used for humans (+Hum) inhabiting a certain place (+Inh) in Switzerland (+CH). This noun is associated with the inflectional class code (N60) which enables automatic production of all of its inflectional forms. For the noun *Ženevljanin* one such automatically produced form is

$$\textit{Ženevljanima}, \textit{Ženevljanin}.N+NProp+Hum+Inh+CH{:}mp3v.$$

Values of associated grammatical categories state that it is a masculine gender (m) animate noun (v) in the plural from (p) in the dative case (3). All markers associated with a lemma are inherited by inflectional forms. Dictionaries of forms automatically produced in this manner are used in NLP tasks.

The rich set of semantic markers is particularly useful in NER tasks. We will discuss these markers in more detail in Section 3. A full list of grammatical categories used in Serbian e-dictionaries and their values as well as an extensive but not exhaustive list of markers is given in Chapter 1 of [14]. For proper names, the system of markers relies on [8].

Among general resources used are also dictionary graphs in the form of FSTs that recognize and grammatically tag certain classes of simple words and compounds, namely those that are generally not to be found in dictionaries because it is not possible to produce a finite list of their canonic forms.

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Examples of simple words of this kind are Roman numerals, interjections with repetition of one or more graphemes (e.g. *jaooo* 'ouuuuch') or acronyms which are not generally known or regularly used. Dictionary graphs are also used to properly tag numerals written with several digits, words or their combination (e.g. *18 milijardi i 800 miliona* '18 billion and 800 million'), compound nouns or adjectives starting with digits (e.g. *21-godišnji* '21 years old'), and 'inflected' forms of acronyms (e.g. *MOK-a* 'the genitive case of MOK—International Olympics Committee') as well as their 'derivational' forms (e.g. *DSS-ovac* 'a member of DSS (political party)'). Some dictionary graphs rely on information in e-dictionaries, e.g. graphs for compound adverbs like *samozadovoljno* 'in self-satisfied manner' derived by prefixation from *zadovoljno* 'in satisfied manner' and *dvadesetostruk* 'twenty times as big' rely on the fact that all or some compound components are already in e-dictionaries. The core set of dictionary graphs for Serbian is described in Chapter 7 of [14].

3 Selection of named entities

The number of named entity types targeted in a NER task can vary from just a few to quite a number of them. For instance, in the MUC-6 task only three tags were introduced: ENAMEX, TIMEX, NUMEX, each with just a few attributes for further refinement [4]. On the other hand, Sekine and Nobata [23] proposed a named entity hierarchy which included as many as 200 different categories and which refined 'standard' categories by introducing subcategories, as e.g. MEASUREMENT for NUMEX, but also introduced many new categories, such as PRODUCT, FACILITY, EVENT, etc. A detailed multilevel taxonomy is incorporated in the Prolex multilingual database [16]. Detailed guidelines about how to tag named entities in texts are given in Chapter 13 of TEI Guidelines P5 [3]. The guidelines provide tags and attributes for names, dates, people and places that enable a very refined description of these basic classes of named entities; no indications are given, however, as to how the named entity tagging is to be performed. A more detailed discussion on named entity recognition and classification is given in [18].

In [22] the authors describe the tools and methods used to produce a linguistic resource, within the National Corpus of Polish, in which named entities are tagged in full detail and with high accuracy. However, our main objective was not to build a similar resource, minutely tagged with named entities in Serbian, which could be used as a kind of a gold standard in the future, but rather to develop a comprehensive, working and useful tool for NER in Serbian newspaper texts. Thus, we chose to tag entities that are considered as basic and hence included in most NER systems. In selecting the set of named entities to be recognized and tagged we also took into consideration the lexical coverage of our electronic dictionaries, as well as the set of semantic markers used in them.

3.1 Numerical expressions

We considered two types of numerical expressions: money expressions and measurement expressions. Under a money or measurement expression we assumed expression consisting of a numeral (written using digits, words or their combination) followed by a currency or a measurement unit. All currencies in use today (and some that are out of use) are in our dictionary (marked with +Cur), as well as major measurement units (marked with +Mes). Besides their full names, currencies and measurement units in numerical expressions can be expressed by acronyms (e.g. USD), abbreviations (e.g. kg) and special symbols (e.g. €).

3.2 Temporal expressions

Two types of temporal expressions were considered: dates and times. In both cases, we distinguished expressions that represent a moment (however vaguely expressed) from those that represent a period

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(*od 1968. godine do danas* 'from the year 1968 until today', *od 11 do 13 sati* 'from 11 to 13 hours'). As for dates, we were not looking only for precisely determined dates as in *13. decembra 2005.* 'on December 13th, 2005' but also for less formal expressions in which the year is omitted and the current year is presumed (*23. novembra* 'on November 23rd'), or the year can be inferred (*15. avgusta prošle godine* 'last year on August 15th'). We were also looking for expressions in which an exact day is not mentioned but the year is explicitly given (*u martu 1999. g.* 'in March 1999'), or the year can be inferred (*u aprilu sledeće godine* 'in April next year'), or the current year is presumed (*za početak novembra* 'for the beginning of November'). Finally, we were looking also for expressions in which only the year is mentioned; however, in that case the word *godina* 'year' had also to appear in full or abbreviated form (*za 2004. godinu* 'for the year 2004'). The names of days were recognized if they were related to a date (*u ponedeljak 2. januara* 'on Monday, January 2nd'), but not if they appeared on their own. Formal expressions of date and time were also recognized, such as *17. IV 2006* or *10:01h*.

3.3 Name expressions

We considered two types of name expressions: geopolitical names and personal names. In recognizing geopolitical names we distinguished four types: names of settlements (*Njujork* 'New York'), names of countries (*Nemačka* 'Germany'), hydronyms (*Dunav* 'Danube', *Atlantski okean* 'Atlantic') and oronyms (*Alpi* 'Alpes'). In recognizing names of countries we were looking for both formal names (*Narodna republika Kina* 'Peoples Republic of China') and names in daily use (*Kina* 'China'). Some frequently used acronyms of countries were also recognized (*SAD* 'USA'). Recognition of these geopolitical names was based on semantic markers in our e-dictionaries: besides the +Top marker, given to all geopolitical names, additional markers are assigned to settlements (+Gr), countries (+Dr), bodies of water (+Hyd) and elevations (+Oro).

Recognition of personal names was also based on semantic markers in our e-dictionaries: namely, the markers +First, +Last and +Nick are assigned to first names, surnames and nicknames, respectively. We recognized full names, i.e., names consisting of a first name and at least one surname, as well as references where only the surname was given. References to persons only by their first name or nickname is very rare except in literary texts or for those very prominent like *Tito* at his time, so we do not see that omitting such references represents a serious drawback. Various titles were also recognized when related to a person's full name (*prof. dr Kata Lazović*). Although a person's function, profession or role in the society is usually not considered as a named entity, we nevertheless recognized them when they directly preceded or followed a personal name, thus forming a nominal phrase: *prof. dr Slavica Đukić-Dejanović, direktor Kliničko-bolničkog centra* 'Prof. Dr Slavica Đukić-Dejanović, director of the Hospital Medical Center' and *Predsednik Odbora za poljoprivredu u Vladi Republike Srbije Jela Veselić* 'the president of the Agricultural Committee in the Government of the Republic of Serbia Jela Veselić'. Special attention was given to recognizing names and positions of distinguished persons, like *papa Benedikt XVI* 'Pope Benedict XVI' and *belgijski kralj Albert drugi* 'Belgian King Albert the Second', because they often follow special naming rules. We think that this adds a significant value to the already recognized personal name.

4 Development of local grammars

Our NER system strongly relies on e-dictionaries of simple words and compounds. In Unitex they are basically used in the following manner: Unitex first creates a dictionary of a processed text as an intersection of the text and e-dictionaries, and then local grammars perform a look-up in the produced

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text dictionary. However, the usage of dictionaries without additional resources cannot result in a successful system due to a high level of ambiguity of both word forms and lemmas. There is an example that illustrates well the nature of this problem. In Serbian, *po* is a very frequently used preposition. The same string with the uppercase initial, *Po*, can represent four different proper names: the Italian river *Po*, the French commune *Pau* (on the northern edge of the Pyrenees), part of a personal name as in *Edgar Allan Poe* and finally the chemical symbol for *Polonium*[3]. However, in newspaper texts these proper names rarely appear. For instance, in one sample of newspaper texts containing 580,000 tokens there were 122 occurrences of the token *Po* and none of them represented any of these proper names. This and many other similar examples suggest that agreement and contextual constraints have to be taken into consideration if we want to obtain results with high recall and precision.

4.1 Personal names

In order to recognize personal names we had to treat in special way:

- Full names (tag <NAME>) with or without their social position (tag <RS>);
- Lists of full names with or without their position in society;
- Names of distinguished persons (tag <VIP> for titles and positions and tag <VNAME> for names);
- Surnames only (tag <LNAME>) with or without their position or title (tag <LRS>).

Full names. Due to the complexity and ambiguity of personal names in Serbian a simple lexical pattern consisting of a first name and surname, as *<N+Hum +First> <N+Hum+Last>* would recognize full personal names with a very low precision. For instance, this search pattern would falsely recognize as a full personal name the sequence *Sofiji Montgomeriju* in the text *…ambasadoru SAD u Sofiji Montgomeriju smetalo je…* '…the US ambassador in Sofia Montgomery was disturbed by…' because *Sofija* is both a female first name and the capital of Bulgaria. The use of simple agreement conditions would, however, reject it. This is explained in full detail in Chapter 6 of [14]. However, precision can be significantly improved without any adverse effect on recall if case-number-gender agreement conditions are taken into account. This supports claims presented in [20] that there are features of Slavic languages, like rich inflection, that can help in some NLP tasks. Thus, e.g., the pattern *<N+Hum+First:ms2> <N+Hum+Last:ms2>* recognizes masculine names (*m*) in the genitive case (*2*) whereas *<N+Hum+First:fs2> <N+Hum+Last:s1>* recognizes feminine names (*f*) in the genitive case. Note that the patterns for masculine and feminine names differ since surname parts in Serbian feminine full names do not inflect. However, local grammars that we built for recognition of personal names are more complex than these simple patterns since they take into account additional features of personal names, such as the optional use of titles, nicknames, an additional surname for women, middle names or initials, as well as the order in which the first name and the surname appear, etc.

Special local grammars were produced for recognizing a person's position in the society that precedes or follows a personal name forming thus a nominal phrase. Such local grammars recognize the following basic forms of phrases:

(1) *generalni sekretar* (*udruženja pravnika Srbije*) gen.phrase—Secretary General of the Lawyers Association of Serbia;

[3]It should be noted that all foreign names in Serbian texts are transcribed, regardless of whether they are written in Latin or Cyrillic alphabet.

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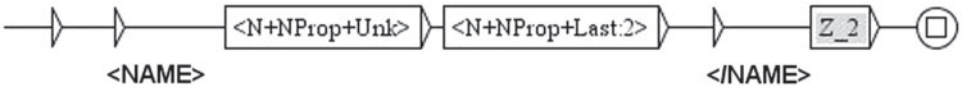

FIGURE 1. The graph that recognizes and tags a partially unknown personal name in the genitive case if followed by a person's position also in the genitive case.

(2) *šef službe* (*za mala i srednja preduzeća*) FOR-prep-phrase—Head of service for small and medium enterprises;
(3) *direktor* (*i većinski vlasnik*) AND-conj-phrase—Director and majority shareholder;
(4) *ministar* (*omladine* (*i sporta*) AND-conj-phrase)gen.phrase—Minister of Youth and Sports;
(5) *ministar prosvete* (*u vladi Republike Srbije*) IN-prep-phrase—Minister of Education in the government of the Republic of Serbia;
(6) *izvršni direktor* (*Beogradske banke AD*abb) gen.phrase—Executive Director of Belgrade Bank AD;
(7) *visoki predstavnik EUabb (za spoljnju politiku i bezbednost) FOR-prep-phrase*—EU High Representative for Foreign and Security Policy;
(8) *predsednik (makedonske Komisije) gen.phrase (za odnose (sa verskim zajednicama) WITH-prep-phrase) FOR-prep-phrase*—President of the Macedonian Commission for Relations with Religious Communities.

These basic structures can be combined in various ways to recognize more complex phrases as demonstrated by the example: <RS><NAME>*Mirjana Dragaš* </NAME>*, predsednik Upravnog odbora Republičkog zavoda za tržište rada i zamenik saveznog ministra za rad i socijalna pitanja*</RS> 'Mirjana Dragaš, Chairman of the Steering Committee of the Republic Institute for Labor Market and Deputy Federal Minister of Labor and Social Affairs'.

We already explained how we used local grammars in order to improve precision in recognition of personal names. These local grammars perform well on names that are in our e-dictionaries, but fail to recognize a full name if one or more of its constituents are unknown. Our e-dictionaries of personal names cover Serbian and English names and only a small number of other foreign names transcribed into Serbian. Consequently, if our local grammars relied on e-dictionaries only, they would fail to recognize most foreign names in Serbian texts, except common English names. For that reason we developed additional local grammars that improve recall of recognition of personal names. These local grammars rely on those already developed that recognize a person's position in the society. Basically they function like this: if a phrase recognized as a (potential) person's position is preceded (or followed) by two unknown words both with uppercase initial or one personal name and one unknown word with uppercase initial then these two words are considered to form a full personal name (Figure 1).

Two examples of successful recognition of personal names that were not in our e-dictionaries are: <RS>*Ministar odbrane Rumunije* <NAME>*Teodor Atanasiu*</NAME></RS> 'Romanian Defence Minister Teodor Atanasiu' (only surname unknown) and <RS><NAME>*Gerasimov Ivanovič*</NAME>*, član kolegijuma saveta međunarodnog saveza pravnika*</RS> 'Gerasimov Ivanovich, a member of the Collegium of the Council of the International Lawyers Association' (both first name and surname unknown).

Lists of names. Lists of full names often occur in newspaper and other non-fiction texts. In these lists names are sometimes preceded or followed by a person's position (<RS>*ministar pravde Haitija* <NAME>*Pol Deni*</NAME></RS> *i* <RS>*opozicioni političar*<NAME>*Mišel Gajar* </NAME></RS> 'Haitian Justice Minister Paul Denis and opposition politician Michel Gaillard'),

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whereas sometimes a single designator is given for all of them (<RS>*Najbolje srpske teniserke* <NAME> *Jelena Janković*</NAME> *i* <NAME>*Ana Ivanović*</NAME></RS> 'The best Serbian tennis players Jelena Janković and Ana Ivanović').

Special treatment of lists is important in order to assign the right position to the right person or persons. The following example illustrates this:

Bilateralne sastanke sa resornim ministrima RS imaće ministri <RS> <NAME> *Aleksandar Popović* </NAME>, *Predrag Bubalo* </RS>, *Bojan Dimitrijević i* <NAME> *Velimir Ilić* </NAME>
Bilateral meetings will be held between competent RS ministers and ministers Aleksandar Popović, Predrag Bubalo, Bojan Dimitrijević and Velimir Ilić

The above result was obtained without special treatment of lists of names, and there are several problems with the tagging: (a) one name (*Bojan Dimitrijević*) is not recognized; (b) the system failed to recognize that all persons mentioned are ministers; (c) a position is attributed to only one person and it is incorrect—<RS><NAME>*Aleksandar Popović*</NAME>*, Predrag Bubalo*</RS> 'Aleksandar Popovic, very dear overzealous student'. This last error occurred due to the homography of forms and the fact that our graphs are always looking for the longest match: the first name *Predrag* is also an adjective 'very dear' and the surname *Bubalo* is also a noun 'overzealous student'. With special treatment of lists of names we obtain the correct tagging:

Bilateralne sastanke sa resornim ministrima RS imaće <RS> *ministri* <NAME> *Aleksandar Popović*</NAME>, <NAME> *Predrag Bubalo*</NAME>, <NAME>*Bojan Dimitrijević*</NAME> *i* <NAME> *Velimir Ilić*</NAME></RS>

Special graphs for the list of names are prepared for the following situations:

- List of known full names without any functions or roles—recognition of a list as a whole solves the problem of 'longest match';
- List of known or potential full names in which each name is either preceded or followed by a function or a role—this solves the problem of correct attribution of a role to a person;
- List of known or potential full names preceded by a common designator that has to be in this case in the plural form. This sub-graph enables recognition of unknown names as in the following example: <RS>*kineski pukovnici* <NAME> *Ćao Lijang*</NAME> *i* <NAME>*Vang Sjangsui* </NAME> </RS> 'Chinese colonels, Cao Liang and Wang Sjangsui'.

Distinguished persons. Distinguished persons, such as members of royal families or aristocracy as well as high-ranking church officials often follow different naming rules. In referring to such persons only their first name is often used, e.g. *car Viljem* 'Emperor William', names used are often specific and usually not in dictionaries, e.g. *kineski car Hsuan Tung* 'Chinese emperor Hsuan Tung' and they can be accompanied by ordinal numbers written in various ways: *britanska kraljica Elizabeta Druga* 'Britain's Queen Elizabeth the Second' and *papa Jovan Pavle II* 'Pope John Paul II'. In naming church officials unusual noun phrases are often used (with reverse order of a noun and an adjective) *vladika osječko-poljski i baranjski Lukijan* 'Bishop of Osijek-Field and Baranja Lucian'. As with regular full names, where part of a person's position can be the name of an organization, such cases are recognized for names of distinguished persons as well:

<VIP>*novoizabrani patrijarh Srpske pravoslavne crkve,* <VNAME> Irinej
</VNAME> *niški.* </VIP>
newly elected Patriarch of the Serbian Orthodox Church, Irenaeus of Nis

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<VIP>*muftija Islamske zajednice Srbije*<VNAME> *Muhamed Jusufspahić* </VNAME> </VIP>
mufti of the Islamic Community of Serbia Muhamed Jusufspahić

Surnames. Recognition of surnames differs substantially from recognition of full names. First of all, ambiguity of surnames is much higher if they are not supported by at least one more name, and the position of a person referred to only by her/his surname is often omitted (usually because it was mentioned before in the text, when the full name was given). In rare situations in which a position is mentioned it is usually only one or two words long, so it cannot be safely used with unknown surnames. Moreover, special forms are used for women referred to by a surname: feminine gender of the possessive adjective—*Milićeva*—or a noun derived from the surname—*Milićka*. Thus, a different strategy was developed for the recognition of surnames only, and it is similar to the one used for geopolitical names (see Subsection 4.2). Local grammars for surnames make intensive use of positive and negative right and left contexts implemented in Unitex (Section 6.3 [19]).

(1) The basic graph recognizes as surnames only simple words that are not ambiguous with any other proper or common name. To that end we use negative right context (the top of Figure 2).
(2) A collection of graphs recognizes an ambiguous surname if it is preceded by a title or a noun describing a person's position. These graphs use negative contexts to exclude unwanted occurrences after a surname, like unknown uppercase words, and ambiguous nouns for positions that could lead to false recognitions (the middle of Figure 2).
(3) A collection of graphs recognizes ambiguous surnames in the nominative case if they represent the subject of a phrase (use of both left and right contexts). To that end a complex module for simple and analytic tenses is used (the bottom of Figure 2).

We can illustrate the results of application of this local grammar with the surname *Koštunica*, which is ambiguous with the common noun *koštunica* 'stone fruit'.

– *Koštunica u Bukureštu* 'Koštunica in Bucharest' (an article heading)—the surname was not recognized, because it is ambiguous and has no appropriate context;
– *Kabinet premijera Koštunice je savetovao...* 'The office of Prime Minister Koštunica has advised...'—the surname was recognized because it is preceded by a position (application of graphs described in item 2);
– *Koštunica je primio delegaciju...* 'Koštunica received a delegation...'—the surname was recognized because it is the subject of a phrase (application of graphs described in item 3);
– *Kritika Koštunice* 'Koštunica critisized' (an article heading)—the surname was recognized but for wrong reasons. The phrase is ambiguous because *kritika* is the nominative case of the noun *kritika* 'critics' and the genitive case of the noun *kritik* 'critic'. The phrase can be interpreted as <LRS>*kritika*<LNAME>*Koštunice*</LNAME></LRS> 'the critic Koštunica (in the genitive case)'. Graphs described in item 2 failed to resolve the ambiguity.

4.2 Geopolitical names

Geopolitical names are ambiguous by themselves, namely, the same name can represent two or more different entities: a city and a country (e.g. *Luksemburg* 'Luxembourg'), a mountain and a region (e.g. *Balkan*), etc., but they can also be ambiguous with other proper names as well as common words. Thus, as in the case of personal names, tagging of geopolitical names cannot rely on e-dictionaries only. An additional problem in Serbian arises from the fact that the names of a number of countries

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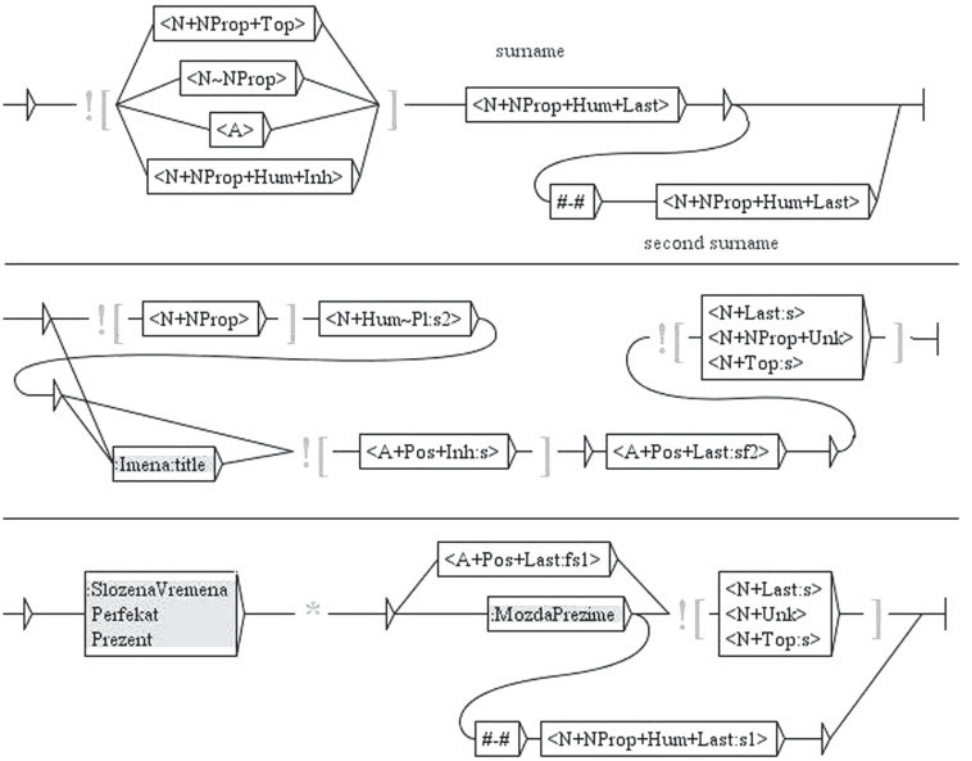

FIGURE 2. Simplified sub-graphs for recognition of surnames.

coincide with the feminine gender relational adjectives used for these names: e.g. *Francuska* 'France' can also mean 'French', and *Hrvatska* both 'Croatia' or 'Croatian'. Thus, various constraints have to be used in order to keep the precision high.

The local grammar for geopolitical names also makes intensive use of positive and negative right and left contexts. We will illustrate how this local grammar functions on the example of the local grammar recognizing names of countries (tag <TOP-C>), which consists of four sub-graphs:

(1) The basic graph recognizes all compound country names, all abbreviated country names, but only those simple word country names that are not ambiguous with any other proper or common name. To that end we use negative right context (the top of Figure 3).
(2) For ambiguous simple word country names that appear in a text in the genitive case we use a graph that recognizes such a name if some trigger noun appears before it, e.g. *predsednik* 'president', *skupština* 'parliament', etc. To that end we use the left context (upper middle of Figure 3).
(3) For simple word country names that are ambiguous with relational adjectives we use a graph that recognizes such a name if it is not followed by a common noun (thus preventing false recognitions in noun phrases like *Francuska banka* 'French bank') or if it is followed by an auxiliary verb (thus allowing correct recognitions in phrases like *Francuska je odlučila* 'France has decided'). To that end we use both positive and negative right contexts (the lower middle of Figure 3).

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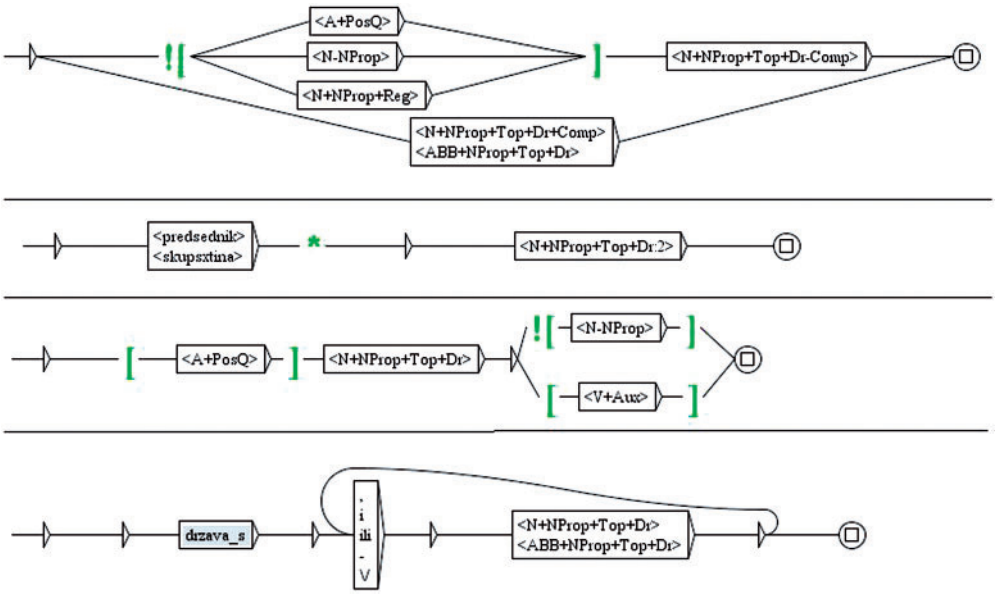

FIGURE 3. Simplified sub-graphs for recognition of country names (Parts that insert XML tags to the processed text are omitted).

(4) Finally, we recognize as a country name every ambiguous simple word country name that appears in some kind of a list with other country names, provided that this list contains at least one unambiguous country name. The bottom of Figure 3 represents the path in a graph that matches a list of names in which the first country name is unambiguous, which is established by the basic graph described in item 1. This path will recognize two country names in *Srbija i Hrvatska* 'Serbia and Croatia', because the first name is unambiguous.

A similar approach is implemented for other types of geographic names: names of settlements (tag <TOP-S>), hydronyms (tag <TOP-W>) and oronyms (tag <TOP-H>). One successfully tagged sentence is:

...prikazani su snimci telekomunikacionog tornja na <TOP-H> *Avali* </TOP-H>*, mostova na* <TOP-W> *Savi* </TOP-W>*,* <TOP-W> *Južnoj Moravi*</TOP-W>*,* <TOP-W> *Ibru* </TOP-W>*, kao i mostova na* <TOP-W> *Dunavu* </TOP-W> *u* <TOP-S> *Novom Sadu* </TOP-S>*.*
Images are shown of the telecommunication tower on Avala, bridges over Sava, South Morava, Ibar, as well as bridges over the Danube in Novi Sad.

4.3 Other named entities

Recognition of money (tag <CURR>) and measurement expressions (tag <UNIT>) also requires intensive use of graphs, particularly for the recognition of the numerical part of the expressions. For that, however, no special graphs were produced, since a system of 40 dictionary graphs for recognition of multi-word numerals was already available and is in regular use for text processing, as described in Section 2. Graphs for numerical expressions rely on information from e-dictionaries, and syntactic structures they have to cover are rather simple.

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Local grammars were also produced for recognition of temporal expressions—dates (tags <DATE-M> and <DATE-P>) and time (tags <TIME-M> and <TIME-P>). These local grammars, similar to those for numerical expressions, use available dictionary graphs for multi-word numerals. As opposed to local grammars for all other named entities, they use information from e-dictionaries to a much lesser degree. However, the expressions they have to recognize come in various different syntactic forms, which makes these local grammars rather complex. The following example illustrates recognition of these types of NEs:

Izgradnja te obilaznice, ukupne dužine <UNIT>*3.600 metara*</UNIT>, *počela je* <DATE-M>*1989. godine*</DATE-M> *i u nju je uloženo* <CURR>*2,5 miliona evra.*</CURR>
Construction of this bypass, with a total length of 3,600 m, began in the year 1989, and 2.5 million euros were invested in it.

Our NER system at present consists of 387 graphs and sub-graphs, many of which are, due to its modular architecture, used in several sub-systems.

5 The experiment and evaluation

For evaluation of the results produced by our local grammars we used a collection of 3,000 short agency news dated from May 2005 to December 2006 that were not used during the development of our NER system. The size of this corpus is ∼155,000 simple word forms, and 5,642 sentences. Thus, an average news item consists of a little less than 2 sentences (1.88), and 51 simple word forms. This collection of news pertains to Serbian internal and external politics.

The local grammars were applied to the corpus in the following order: measurement expressions, money expressions, dates, personal names and roles, time of day, geopolitical names. This order is the simple consequence of the fact that the local grammars were applied in the order in which they were actually produced. Being last produced, local grammars for surnames and distinguished persons did not enter this evaluation, and they were evaluated separately at a later stage. Each tagged text was carefully verified by one human evaluator, and all inserted tags were checked. This process was repeated twice on the same collection of texts, and the results of the first round of evaluation were used to make many improvements and amendments to our NER system. The application of our improved NER system to the same collection of texts enabled us to effectively measure the achieved improvements.

The evaluators also inserted a new attribute CHECK into every tag with the value 'OK' if the named entity was correctly recognized and tagged, or 'NOK' if this was not the case or if it was only partially recognized. For partially correct tagging the assigned attribute value was also 'NOK', as for instance in the following case, where the last part of a personal name was not recognized: …*izjavio je* <RS CHECK='OK'>*šef španske diplomatije* <NAME CHECK='NOK'>*Migel Anhel*</NAME></RS> *Moratinos*…'…said the head of Spanish diplomacy Miguel Angel Morations…'. If the named entity was totally missed, the appropriate tag with the value 'MISS' in the check attribute was inserted. One such example is: …*od* <RS CHECK='MISS'>*generalnog sekretara NATO* <NAME CHECK='MISS'>*Jap de Hop Shefera* </NAME> </RS> …'by NATO Secretary General Jaap de Hoop Scheffer…'.

It was not always easy to decide which value for the check attribute CHECK is the most appropriate. Since our corpus dated from years 2005 to 2006 when the State Union of Serbia and Crna Gora ceased to exist, it was not always easy to determine whether occurrence of *Srbija i Crna Gora* in the corpus represents one or two named entities. It should be noted, however, that we at present neglect the fuzzy nature of some named entities and always treat as correct, for instance, geographic place tags assigned to names of settlements, although their actual usage could belong to another category, like

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TABLE 2. Results of the second round of evaluation

Named entities	OK		NOK		MISS		Token/ type	Precision		Recall		F-measure	
	Token	Type	Token	Type	Token	Type		Token	Type	Token	Type	Token	Type
TIME-P	8	8	0	0	0	0	1.0	1.00	1.00	1.00	1.00	1.00	1.00
TOP-C	3,768	353	19	8	18	15	10.3	0.99	0.98	1.00	0.96	1.00	0.97
CURR	313	230	3	3	4	4	1.4	0.99	0.99	0.99	0.98	0.99	0.99
DATE-M	506	369	10	10	3	3	1.4	0.98	0.97	0.99	0.99	0.99	0.98
TOP-H	24	12	1	1	0	0	2.0	0.96	0.92	1.00	1.00	0.98	0.96
UNIT	281	136	4	4	8	8	2.0	0.99	0.97	0.97	0.94	0.98	0.96
DATE-P	104	84	2	2	3	3	1.2	0.98	0.98	0.97	0.97	0.98	0.97
TOP-S	3,660	666	49	30	163	112	4.9	0.99	0.96	0.96	0.86	0.97	0.90
TIME-M	71	66	4	4	1	1	1.1	0.95	0.94	0.99	0.99	0.97	0.96
NAME	3,344	1,539	94	77	532	347	2.1	0.97	0.95	0.86	0.82	0.91	0.88
TOP-W	16	13	0	0	5	3	1.3	1.00	1.00	0.76	0.81	0.86	0.90
TOTAL	12,095	3,476	186	139	737	496	3.2	0.98	0.96	0.94	0.88	0.96	0.92
RS	2,335	1,260	140	68	407	347	1.7	0.94	0.95	0.85	0.78	0.90	0.86

organization. For instance, it is clear that in *razgovori Beograda i Prištine* 'talks between Belgrade and Priština', Belgrade and Priština represent government institutions rather than locations. The authors in [15] describe the NER approach for French that was built upon large-scale dictionary resources and graphs which use the wider context to resolve such ambiguities. Their approach relies on a thorough description of phrasal verbs that does not yet exist for Serbian.

The results of our experiment after the second round are summarized in Table 2. The sample we used contained 12,832 NE tokens and 3,972 NE types, hence 3.2 tokens per type. The most frequent tokens are personal names (NAME), followed closely by names of settlements (TOP-S) and names of countries (TOP-C), with a total of 3,876, 3,823 and 3,786, respectively. These three categories account for 89.5% of the total number of NE tokens. As for NE types, almost half of them represent personal names (47.48%), and another 19.59% are settlements. With 9.26% for countries, these three categories account for 76.33% of total NE types. The highest token/type ratio is 10.3 for names of countries, and the lowest, one token per type, for temporal expressions representing time periods (TIME-P), which does not come as a surprise.

If we look at NE tokens only, the precision of our NER system is 0.98 whereas its recall is 0.94, yielding an overall F-measure of 0.96. The highest precision of 1.0 was reached for time periods and names of bodies of water (TOP-W), followed by names of countries, and money (CURR) and measurement (UNIT) expressions, all with a precision of 0.99. On the other hand, the lowest precision of 0.95 was achieved for expressions for time periods, followed by 0.96 for names of oronyms (TOP-H). However, names of oronyms have a recall of 1.0, hence an overall F-measure of 0.98. The lowest recall of only 0.76 was reached by names for bodies of water, followed by personal names with a recall 0.86. As a consequence, named entities representing bodies of water have also the lowest F-measure of 0.86 followed by personal names with 0.91. On the other end are names of countries and expressions for time periods, which have an F-measure of as much as 1.00.

If we look at NE types, the results do not differ very much. The overall precision is 0.96, the recall 0.88 and the F-measure 0.92. The highest possible precision of 1.0 goes once again to expressions for time periods and names of countries, but this time closely followed by currencies with 0.99. Oronyms are at the bottom of the precision list with a score of 0.92. Named entities representing bodies of water and personal names have the lowest recall of 0.81 and 0.92 as well as the lowest F-measure of

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TABLE 3. Evaluation results for distinguished persons and surnames

Named entities	OK		NOK		MISS		Token/ type	Precision		Recall		F-measure	
	Token	Type	Token	Type	Token	Type		Token	Type	Token	Type	Token	Type
VNAME	44	25	2	2	0	0	1.8	0.96	0.93	1.00	1.00	0.98	0.96
VIP	39	30	5	5	1	1	1.3	0.89	0.86	0.98	0.97	0.93	0.91
LRS	43	23	8	8	9	8	1.7	0.84	0.74	0.83	0.74	0.83	0.74
LNAME	788	334	117	74	327	156	2.3	0.87	0.82	0.71	0.68	0.78	0.74

0.90 (the same as settlements) and 0.88, respectively. At the top of the F-measure list are expressions for time periods with 1.00 closely followed by money expressions with 0.99.

Finally, we would like to mention also the success rate of the recognition of a person's position in the society (RS). As these are not usually considered as NEs, they are represented separately in the bottommost row of the general overview table. However, for 2,742 tokens and 1,607 types related to such expressions, the precision was 0.94 and 0.95, respectively, with a recall of 0.85 for tokens and 0.78 for types, and an F-measure of 0.90 and 0.86. Given the complexity and variety of such expressions this can be perceived as a very successful outcome.

We performed an additional evaluation in which we were specifically interested in the efficiency of our local grammars developed for recognizing names of distinguished persons and local grammars recognizing surnames, with or without their position in the society. To that end we used a subset of the same collection of agency news (1,800 pieces of news), to which we applied our local grammars and once again manually checked and labelled the results. An overview of these results is given in Table 3. The vast majority of both the tokens and types recognized are surnames (LNAME) (89.13% and 84.92%, respectively), with a token/type ratio of 2.3. Although the precision of the local grammars for surname tokens and types is reasonably good (0.87 and 0.82, respectively), the recall is not (0.71 and 0.68), and neither are the resulting F-measures of 0.78 and 0.74.

The recognition of the position of the person (LRS), when it accompanied his/her surname, yielded balanced but also moderate results for precision and recall, albeit better for tokens than for types. When compared with the evaluation results for the first set of named entities summarized in Table 2, the performance of local grammars for surnames is considerably lower. However, one must bear in mind that the results presented for the first set were obtained after a second round of evaluation, which came after an improvement of local grammars based on the evaluation results of the first round. We believe that the same can be done for local grammars for surnames, resulting in improved performance, and it is indeed what we plan to do in the nearest future.

The results for recognition of names (VNAME) and positions (VIP) of distinguished persons are much better, especially in terms of recall (1.00 for name tokens and types, 0.98 and 0.97 for tokens and types pertaining to the position), yielding very good F-measures for all cases. However, one must take these results with a grain of salt given the relatively small size of the sample.

In general, the results of this additional experiment are encouraging. However, they also show that we need to put some more effort in enhancing our local grammars that recognize surnames. Finally, this evaluation confirmed the fact that was already obvious in the initial evaluation, namely, that our system gives priority to precision.

The analysis of the obtained results showed that the causes of omissions and incorrect tagging were various and can be classified as follows:

- *Typographic, orthographic and grammatical errors in the source text*. This happened quite often with named entities representing numerical expressions because authors tend to use a

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decimal point instead of a decimal comma required by the Serbian orthography (*4.4 stepena Rihterove skale* '4.4 on the Richter scale' should have been written as *4,4 stepena Rihterove skale*). Also, some authors do not abide to agreement rules specific for numerical phrases: in *svega četiri metara* 'just four meters' the measurement expression was not recognized because it was not written correctly as *četiri metra*. At this moment we decided not to deal with failures due to incorrect input.

- *Recognition in titles*. Our system does not perform well on article titles written in all upper case, because in that case it is not easy to distinguish between proper and common names. For instance, false recognition <NAME CHECK='NOK'>VLADA MORA</NAME>DA ZAPOSLI GRADJANE would not occur if this title was written in the regular form: *Vlada mora da zaposli gradjane* 'The government has to employ the citizens'.
- *Absence of a name in e-dictionaries.* Problems of this type are often caused by foreign names. As presented in Section 4, we tried to deal with this problem but not at the cost of lowering the precision. Foreign surnames when used on their own tend to have a rather small recall, which cannot be improved easily because they are often ambiguous with Serbian common names.[4] The fact that foreign names are always transcribed into Serbian, that transcription rules are sometimes difficult to understand and are also changing quite often additionally complicates this problem.[5]
- *Oversights and minor deficiencies in the construction of local grammars*. This happened more often for named entities for which we had not enough data for modelling, like geographic names for hydronyms and oronyms and names of distinguished persons.[6] Also, our local grammars failed in cases of some complex syntactic constructions like nested structures, e.g. *portparol glavnog tužioca Haškog tribunala Karle del Ponte Florans Artman* 'spokesman for Chief Prosecutor of Hague Tribunal Carla del Ponte, Florence Hartmann'.

6 Future work

Although we see the evaluation results of our NER system as promising, there is still much to be done. Besides improving the existing system on the basis of evaluation results, our future work will concentrate on enhancing it to recognize more NE categories. The modules for organization names and some urban names were already developed on principles similar to those described in this article and their evaluation is in progress. The performance of our NER system is being further improved by organizing graphs in cascades, as suggested in [7], which does not only facilitate development but also partially solves the problem of recognition of embedded nominal phrases (the problem discussed in [6]). For instance, the problem illustrated by the example from the last item of the

[4]In our corpus, due to the time period it covers, names of *Havijer Solana* 'Javier Solana', *Oli Ren* 'Olli Rehn' and *Doris Pak* 'Doris Pack' appeared quite often, and as Serbian readers were familiar with them, they were often referred to by surnames only. All three surnames are ambiguous with Serbian common names: *solana* 'saltpan', *ren* 'horse radish' and *pak* 'puck'.

[5]A recent processing of Jules Verne's novel 'Five weeks in a balloon' translated to Serbian in 1970s revealed that transcription for almost all geographic names of places situated in today's Tanzania or near-by regions changed: the island *Bemba* is today transcribed as *Pemba* 'Pemba', the region *Muanza* became *Mvanza* 'Mwanza', the lake *Ukerjue* became *Ukereve* 'Lake Victoria', etc. The list is endless.

[6]It is interesting to note that although both names of members of royal families and names of church officials appear in newspaper texts rarely, their distribution changes in time. In a small diachronic corpus of newspaper texts (a 100 years span) of 55,000 simple words, 53 references to kings and aristocracy were retrieved, and only 4 names of church officials, while in a small contemporary newspaper text of 100,000 simple words only 3 references to kings were retrieved with as many as 16 church officials.

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previous section can be solved by this new organization. After inclusion of remaining NEs into our system (events, products, inhabitants, etc.) we will move from NE recognition to IE by tackling the problem of normalizing the recognized NEs.

Acknowledgements

We would like to thank MSc and PhD students at the Faculty of Philology, University of Belgrade for their help in producing some of the modules used in the NER system for Serbian: Sandra Gucul Milojević, Vanja Radulović, Jelena Jaćimović and Bojana Đorđević. We are also grateful to students of Library and Information Sciences at the same faculty who performed the evaluation task within the scope of the courses 'Information Retrieval' and 'Advanced Information Retrieval'. This research was supported by the Serbian Ministry of Education and Science under the grant #III 47003.

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